

MOBINA NOORANI

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SUMMARY

AI/ML researcher and engineer specializing in agentic AI, reinforcement-learning post-training, multimodal reasoning, retrieval-augmented systems, and computer vision. Build end-to-end AI systems spanning data pipelines, tool-using agents, model fine-tuning, evaluation, and distributed multi-GPU training on HPC infrastructure. Research record includes Scientific Reports and a peer-reviewed NeurIPS workshop publication, with experience translating AI research into robust real-world systems.

TECHNICAL SKILLS

Programming: Python, SQL, JavaScript, HTML/CSS, PHP

ML / Deep Learning: PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, OpenCV

AI Systems: Agentic AI, RAG, multimodal learning, vision-language systems, LLM/VLM fine-tuning, reinforcement learning, GRPO, reward modeling, tool use, self-verification

Infrastructure: Multi-GPU HPC, SLURM, CUDA, FSDP, Ray, FAISS, distributed training, FlashAttention

Tools: Git, Jupyter, Weights & Biases, MySQL, REST APIs

EXPERIENCE

Graduate Research Assistant | University of Florida | Gainesville, FL

Sep 2023 - Present

- Develop agentic multimodal AI pipelines for real-world visual reasoning, retrieval, and automated decision support across large-scale imagery and knowledge-intensive tasks.
- Train vision-language systems with reinforcement-learning post-training to decide when to retrieve external knowledge, reason over evidence, and self-verify answers for knowledge-intensive visual question answering.
- Developed RepSC-SOM, a spatially constrained self-organizing map with representative initialization, adaptive geographic filtering, and region-growing refinement; achieved stronger contamination-hotspot separation than standard geographic units in a published study.
- Built an agentic planning-support framework combining language-model reasoning with human-in-the-loop interaction for disaster-adaptation regionalization.

Teaching Assistant | University of Florida | Gainesville, FL

May 2024 - Aug 2025

- Mentored students in Python, data preprocessing, machine learning, and deep-learning implementation across course datasets.

Teaching & Research Assistant | Alfred University | Alfred, NY

Jan 2023 - Aug 2023

- Developed and tested quantum machine-learning algorithms and mentored 30+ students in programming fundamentals and implementation.

Web & Backend Developer | Pardis Tehran Science & Technology Park | Tehran, Iran

Nov 2018 - Dec 2019

- Designed and shipped full-stack web applications using PHP, JavaScript, MySQL, REST APIs, and Git-based team workflows.

SELECTED PROJECTS

Knowledge-Intensive VQA with Reinforcement Learning & Retrieval

2025 - Present

- Designed a multi-turn agentic pipeline that learns when and how to search external knowledge using outcome-based reinforcement learning and large-scale retrieval.
- Built retrieval over 21M Wikipedia passages and scaled training/evaluation on multi-GPU HPC with distributed training, experiment tracking, and reproducible evaluation workflows.

Multimodal Flood-Risk Vision Systems

2026 - Present

- Develop object-detection and vision-language methods to identify flood-relevant architectural features from satellite and street-view imagery for vulnerability assessment.

Digital Twin / RepSC-SOM Regionalization

2024 - 2025

- Developed a SOM-based regionalization method and applied it to flood-induced E. coli contamination data in Jacksonville, enabling more precise hotspot delineation than conventional geographic units.

Agentic AI Planning Support System

2025

- Created an interactive AI framework for natural-language-guided spatial planning, constraint exploration, and human-in-the-loop refinement of adaptation regions.

EDUCATION

PhD in Artificial Intelligence—Urban planning | University of Florida | Gainesville, FL

Expected 2027

Research focus: Agentic AI, multimodal learning, computer vision, knowledge-grounded reasoning, reinforcement learning, and intelligent systems.

M.Sc. in Electrical & Computer Engineering University of Florida Gainesville, FL	2026
M.Sc. in Artificial Intelligence Malek Ashtar University Tehran, Iran	2021
B.Sc. in Information Technology University of Semnan Semnan, Iran	2017

SELECTED PUBLICATIONS

- Knowledge-Intensive Visual Question Answering with Agentic Retrieval and Reinforcement Learning. Mobina Noorani et al. Manuscript in preparation, 2026.
- Demand-Oriented Regionalization with Local Data. Noorani, M., Gao, S., Chen, C., Saldana Ochoa, K. Scientific Reports (Nature Portfolio), 2026.
- Enhancing Demand-Oriented Regionalization with Agentic AI and Local Heterogeneous Data for Adaptation Planning. Noorani, M., Gao, S., Chen, C., Saldana Ochoa, K. NeurIPS 2025 UrbanAI Workshop (peer-reviewed), 2025.
- Promising Applications for Promoting Disaster Resilience on College Campuses. Noorani, M., Chen, C., Smith, K., Fernandez, F., Saldana Ochoa, K. The Digitized Campus: AI and Big Data in Higher Education, SUNY Press, 2026.

SELECTED PROFESSIONAL DEVELOPMENT

- Transformers in Practice | Sharon Zhou, DeepLearning.AI / AMD In Progress
- Fine-tuning & RL for LLMs: Intro to Post-training | Sharon Zhou, DeepLearning.AI / AMD In Progress
- Post-training of LLMs | Banghua Zhu, DeepLearning.AI / NexusFlow In Progress
- Reinforcement Fine-Tuning LLMs With GRPO | Travis Addair & Arnav Garg, DeepLearning.AI / Predibase In Progress
- Agentic AI | Andrew Ng, DeepLearning.AI Nov 2025
- Retrieval-Augmented Generation (RAG) | DeepLearning.AI Dec 2025
- Generative Adversarial Networks Specialization | Sharon Zhou, Coursera Completed
- Generative AI with Large Language Models | Chris Fregly, Coursera Completed
- Custom Models, Layers & Loss Functions with TensorFlow | Laurence Moroney, Coursera Apr 2022
- Convolutional Neural Networks | Andrew Ng, Coursera Apr 2021
- Structuring Machine Learning Projects | Andrew Ng, Coursera Aug 2020
- Improving Deep Neural Networks | Andrew Ng, Coursera Jul 2020
- Neural Networks and Deep Learning | Andrew Ng, Coursera May 2020